

How each objective supervises the Jacobian

MSE vs. flow/MIP/score-based vs. hetero-Gaussian/Student- t — derivations, with claims

tagged [\[derived\]](#) / [\[measured\]](#) / [\[open\]](#)

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Notation. Observation $x \in \mathbb{R}^d$ (normalized window), action chunk $y \in \mathbb{R}^m$. Expert conditional $p(y|x)$ with mean $\mu(x)$; on clean data $p(y|x) = \delta(y - \mu(x))$. Network f_θ ; encoder Jacobian $J_x = \partial f / \partial x$. Data measure $\rho(x)$ over a *finite* set of windows. “Stiffness” means the curvature of the population loss at its optimum — the restoring force resisting parameter drift; its relevance comes from the SGD stationary-fluctuation relation (§1.3), which converts (stiffness, gradient noise) into (retained, eroded).

1 MSE: the Jacobian is only supervised through finite differences

$$L_{\text{MSE}} = \mathbb{E}_{x,y} \|f(x) - y\|^2, \quad f^* = \mu. \quad (1)$$

1.1 Value-space curvature contains no derivative term

[\[derived\]](#)

Perturb $f \rightarrow f^* + \varepsilon g$:

$$\Delta L = \varepsilon^2 \mathbb{E}_\rho \|g(x)\|^2. \quad (2)$$

Curvature in direction g depends *only on the values of g at the data points*. A perturbation vanishing on the N data points but with a large gradient between them costs exactly zero population loss: with finite data, such perturbations form an infinite-dimensional flat manifold, and nothing in the objective selects among Jacobians consistent with the values.

1.2 The implicit stiffness on J_x

[\[derived\]](#)

The Jacobian is pinned only through *pairs* of nearby samples. Take x and a neighbor $x' = x + \delta v$ ($\|v\| = 1$) with labels $\mu(x), \mu(x')$. Fitting both forces the directional derivative $J_x v \approx (\mu(x') - \mu(x))/\delta$. If the network carries $J_x v = D$ instead, the value error at x' is $\delta \|D - \partial_v \mu\| + O(\delta^2)$, so aggregating neighbors, the quadratic form on the Jacobian error $E = J_x - \partial \mu / \partial x$ is

$$\Delta L \approx \mathbb{E}_{x' \sim \rho_{\text{near } x}} \|E(x' - x)\|^2 = \text{tr}(E \Sigma_\rho(x) E^\top), \quad (3)$$

with $\Sigma_\rho(x)$ the *local input covariance of the data* around x . Consequences: (i) the *target* for J_x is $\partial \mu / \partial x$ — in a label-poor region (μ locally constant) MSE *demand*s a degenerate Jacobian; (ii) the *stiffness* is Σ_ρ -weighted: directions the data does not locally vary in, or regions rarely visited, get near-zero restoring curvature. With label noise σ_n^2 on the pair, the finite-difference estimate of $\partial_v \mu$ has variance σ_n^2 / δ^2 : **the gradient signal on J_x has SNR $\delta |\partial_v \mu| / \sigma_n$** — small label variation under noise makes the Jacobian channel signal-free.

1.3 Erosion corollary

[\[derived\]](#) + [\[measured\]](#)

Model SGD along a curvature eigendirection with stiffness κ , per-step gradient-noise power s^2 , learning rate η (Ornstein–Uhlenbeck):

$$\text{Var}_\infty(\theta) \approx \frac{\eta s^2}{2\kappa}. \quad (4)$$

For MSE at convergence on data containing an irreducible residual mass (aliased switch timing / aleatoric noise), $s^2 = O(1)$ *forever*, while κ on fine distinctions is the tiny Σ_ρ -weighted quantity of Eq. (3). The stationary excursion exceeds the distinction scale $\Rightarrow$ erased; the surviving Jacobian concentrates on the few directions with large label-variation stiffness. Measured: settle-metric PR $13.4 \rightarrow 2.0$ with $s_{\max}/s_{\text{med}} \rightarrow 61$ on pointing data; absent on MP data ($s^2 \rightarrow 0$: PR arrests at 5.1); reproduced with visual controls in the erosion toy.

2 Flow matching / MIP / score-based: the Jacobian is an explicit target

2.1 Conditional targets and their Jacobians

[derived]

Interpolant $y_t = (1 - t)\epsilon + ty$, $\epsilon \sim \mathcal{N}(0, I_m)$, $t \sim U(0, 1)$. For the deterministic conditional $y = \mu(x)$, given (x, y_t, t) one can invert $\epsilon = (y_t - t\mu)/(1 - t)$, so every standard regression target is a *closed-form function* of (x, y_t, t) :

parameterization	target	$\partial(\cdot)/\partial y_t$	$\partial(\cdot)/\partial x$
velocity ($v = y - \epsilon$)	$v^* = \frac{\mu(x) - y_t}{1 - t}$	$-\frac{1}{1 - t} I_m$	$\frac{1}{1 - t} \frac{\partial \mu}{\partial x}$
ϵ -prediction	$\epsilon^* = \frac{y_t - t\mu(x)}{1 - t}$	$\frac{1}{1 - t} I_m$	$-\frac{t}{1 - t} \frac{\partial \mu}{\partial x}$
score $\nabla_{y_t} \log p_t$	$s^* = -\frac{y_t - t\mu(x)}{(1 - t)^2}$	$-\frac{1}{(1 - t)^2} I_m$	$\frac{t}{(1 - t)^2} \frac{\partial \mu}{\partial x}$

Two structural facts shared by all three parameterizations:

(A) A full-rank isotropic Jacobian block is explicitly demanded. The target's y_t -derivative is $\pm I_m/(1 - t)^k$ — rank m , isotropic, at *every* (x, t) . Unlike §1.1 this is not an implicit finite-difference constraint: the loss input itself sweeps a full-dimensional Gaussian ball of y_t around $t\mu(x)$ (radius $1 - t$), so value constraints on the ball pin the derivative *within* it by direct integration — the network must hold m independent unit-scale (after the $(1 - t)$ factor) sensitivity directions everywhere on the tube. Toy verification: $(1 - t) \partial v / \partial y_t = -1.02 \pm 0.25$ vs. the analytic -1 [measured].

(B) The x -Jacobian is supervised with amplification $t/(1 - t)^k$. A network error $\delta\mu(x)$ in the transmitted state-dependence costs $\|\delta\mu\|^2/(1 - t)^2$ at time t (velocity form), so integrated over $t \sim U(0, 1)$, the stiffness on every state-discriminating direction that μ uses is

$$\kappa_{\text{FM}}(x, v) \propto \left(\int_0^1 \frac{dt}{(1 - t)^2} \right) \rho(x) \|\partial_v \mu\|^2 \gg \kappa_{\text{MSE}}(x, v), \quad (5)$$

the integral cut off at the smallest retained noise scale (or by time discretization). Note the scope: the stiffness *per unit of true label variation* is amplified; a direction with $\partial_v \mu = 0$ still has target zero. Flow does not invent rank the task lacks — it prevents the *pruning* of rank the task has, by making maintenance curvature large relative to SGD noise (Eq. (4) with boosted κ).

2.2 Supervision over a measure: no memorization fixed point

[derived] +

[measured]

MSE's supervision set is the finite list $\{(x_i, \mu_i)\}$: memorizing it gives zero loss and zero gradient — from then on the noise of §1.3 is unopposed. FM's supervision set is $\{(x_i, y_t, t)\}$ with fresh (ϵ, t) each step: a *continuous measure*. No finite parameter vector reaches zero loss on unseen draws unless the function is correct as a function on the tube; the maintaining gradient never vanishes. Experimental form: the frozen-noise control — freezing (ϵ_i, t_i) per sample (identical objective, finite list) restored memorize-then-drift dynamics and destroyed the readout, while fresh draws held fine structure indefinitely [measured].

2.3 Predictions vs. what stays empirical

(i) Encoder-rank lift with a dataset-independent signature (the stiffness in 2.1B is objective-supplied) [\[derived\]](#)→[\[measured\]](#): MIP PR $\approx 5\text{--}7$, k_{90} $2\times$ L2's, anisotropy halved, on pointing, MP-200, MP-2k, and human data. (ii) Arrest level tracks the objective, not demo count [\[measured\]](#): MIP 4.2/4.8/5.2 across 200/2k/20k. (iii) **Off-support behavior is not derivable from the objective**: the tube supervises action-space perturbations at on-support x ; state-space extrapolation inherits only the trunk's smoothness bias. The restoring annulus field ($\overline{\Delta d} < 0$), excursion capping (5% vs. 28% escalation), and off-manifold $\|J_x\|$ suppression (edgejac $1.4\text{--}2.6\times$) are measured, consistent with but not implied by §2.1 [\[measured\]](#)/[\[open\]](#).

3 Hetero-Gaussian NLL: the same implicit channel, re-priced

$$L_{\text{HG}} = \mathbb{E} \left[\frac{\|f(x) - y\|^2}{2\sigma(x)^2} + m \log \sigma(x) \right]. \quad (6)$$

3.1 σ -stationarity

[\[derived\]](#)

At the optimum in σ (holding f): $\sigma^2(x) = \frac{1}{m} \mathbb{E}[\|y - f(x)\|^2 | x]$ — the local mean-square residual. Substituting back, the gradient on f is $(y - f)/\sigma^2(x)$: **per-state equalized** — every state contributes $O(1)$ pressure regardless of residual scale.

3.2 Jacobian stiffness: MSE's formula times $1/\sigma^2$

[\[derived\]](#)

HG is a value-space loss, so Eq. (3) applies with the weight carried through:

$$\kappa_{\text{HG}}(x, v) = \frac{\kappa_{\text{MSE}}(x, v)}{\sigma^2(x)}. \quad (7)$$

In the OU relation: in a fitted low-noise region $\sigma^2 \rightarrow \sigma_{\min}^2$ (floor), stiffness amplified up to $1/\sigma_{\min}^2$; in a noisy region $\sigma^2 \rightarrow$ the large irreducible residual, so the gradient-noise power that region exports is damped by $1/\sigma^4$ in variance. **Both numerator and denominator of Eq. (4) move favorably**, hence arrest-and-hold: scripted pointing data, HG arrests at PR 6.5 and holds 200k+ steps while L2 slides to 2.0 [\[measured\]](#); the constructive proof that weights alone carry it: **snorm** ($w = 1/(m + \sigma_{\min}^2)$), plain weighted MSE, no likelihood) reaches SR 93 on true scripted data [\[measured\]](#).

3.3 Why HG's chart can exceed everyone's on noisy data [\[derived\]](#) candidate + [\[measured\]](#)

On heavy-tailed data the HG gradient $(y - f)/\sigma^2$ is *unbounded in y* : a tail draw at 10σ exerts $10\times$ force. Moving f at x_i without disturbing nearby x_j requires a high-gain discriminating direction (§1.2 with $|\Delta y|$ large): outlier-chasing is *rank-building pressure*. Measured signature: HG's human chart is the richest of any arm (PR 11.4, k_{90} 21) while its μ placement is the most corrupted (worst AS= 1 cell, volatile bands) [\[measured\]](#). The chart-corruption link is a candidate, not derived; competing channel: the NLL can absorb an outlier by inflating $\sigma(x)$ locally at only logarithmic cost — which channel dominates depends on the σ -head's smoothness [\[open\]](#).

4 Student- t NLL: same amplification where fitted, bounded influence where not

$$L_{\text{HT}} = \mathbb{E} \left[\frac{\nu+m}{2} \log \left(1 + \frac{\|y-f(x)\|^2}{\nu\sigma(x)^2} \right) + m \log \sigma(x) \right], \quad \frac{\partial L}{\partial f} = - \underbrace{\frac{\nu+m}{\nu\sigma^2 + \|r\|^2}}_{w(r)} r, \quad r = y - f. \quad (8)$$

4.1 Two regimes of the effective weight

[derived]

Small residuals ($\|r\|^2 \ll \nu\sigma^2$): $w \rightarrow (\nu + m)/(\nu\sigma^2)$ — the same $1/\sigma^2$ re-pricing as HG. On clean scripted data all residuals sit at the floor, so HT and HG are gradient-identical up to a constant — their metric ladders coincide to two decimals (6.46/6.47) and both boundaries hold [measured]. **Large residuals** ($\|r\|^2 \gg \nu\sigma^2$): $w \propto 1/\|r\|^2$, so the per-sample force $w r \propto r/\|r\|^2$ — bounded influence, decaying in outlier size: the tail mass exerts vanishing pressure on f , and therefore (by §1.2) no rank-building pressure either.

4.2 Consequences by regime

[derived] + [measured]

Scripted side: HT = HG (regime 1 everywhere); the tail is inert, σ -repricing carries the arrest [measured]. Human side: HT declines to chase the tail (regime 2 on heavy draws), so its chart stays L2-like (PR 5.2 vs. HG 11.4) while its μ is the robust location of the conditional (the t -MLE), on the central mode instead of the tail-dragged mean. Its SR dominance (0.90, flat over AS) with a plain spectrum is the strongest evidence that on noisy data the operative quantity is μ -placement, not the Jacobian spectrum [measured]. Duality note: $w(r)$ is exactly the weight that bounds the influence function of the location estimate — loss tail $\leftrightarrow$ noise tail (Student- t NLL = MLE under the measured $t(2)$ demo noise).

5 Summary

objective	stiffness $\kappa(x, v)$ on J_x	supervision set	noise-mass export s^2
MSE	$\rho(x) \lambda_{\Sigma_\rho}(v)$, target $\partial_v \mu$	finite list	$O(1)$, undamped
flow/MIP/score	$\times \int \frac{dt}{(1-t)^2}$ amplification + explicit full-rank y_t -block $I_m/(1-t)^k$	continuous measure (no zero-loss fixed point)	absorbed by anchor/ ϵ -channel
hetero-G	$\kappa_{\text{MSE}}/\sigma^2(x)$, $\sigma^2 \rightarrow$ local residual	finite list	damped $\times 1/\sigma^4$
hetero- t	= HG for $\ r\ \ll \sqrt{\nu}\sigma$; $w \propto 1/\ r\ ^2$ beyond	finite list	bounded per sample

Unifying statement. Every objective that preserves the Jacobian does so by keeping the maintenance curvature large relative to gradient noise in the SGD stationary balance $\text{Var}_\infty = \eta s^2/2\kappa$: flow by amplifying κ through the $1/(1-t)$ schedule *and* refreshing the supervision measure so the gradient never dies; the hetero-NLLs by re-pricing $\kappa \rightarrow \kappa/\sigma^2$ *and* damping s^2 from the noise mass; MSE does neither, and its Jacobian survives only where the data itself supplies curvature.

Open items for derivation (ranked): (1) off-support extension — why tube supervision at on-support x yields the measured restoring annulus field and $\|J_x\|$ suppression (candidate: the ϵ -channel forces f toward a contraction onto the action manifold in y_t ; coupling to x -extrapolation through the shared trunk’s Lipschitz budget); (2) the HG chart-building vs. σ -absorption competition (§3.3); (3) the human-data 10^6 Frobenius concentration in L2/HT/MIP but not HG (zero-variance input dims) [measured], unexplained.